Introduction

RYH (Ring your health) is a smart ring hardware and an AI-powered companion app for kids and teens aged 10-18 and their parents aged 30-45. The ring tracks sleep, HRV, SpO_2, resting heart rate, skin temperature, and physical activity, converting metrics into plain-English, age-appropriate insights. Unlike adult rings such as Oura or Ultrahuman, RYH operates across two modes: Junior Mode (10–14) with broader parental oversight, and Teen Mode (14–18) giving teenagers control over what data they share.

It is aligned with UN Sustainable Development Goal 3, specifically Target 3.4, and is built around calm, non-judgmental health communication designed to reduce tracking-induced stress rather than cause it.

The plan is supported by secondary research (Google Trends, UK market reports, peer-reviewed literature) and primary research: qualitative interviews testing the concept (Christensen et al., 2016), a Google Form survey (n=40), and website prototype testing via Microsoft Clarity. Five-year Google Trends data confirms sustained rising 'smart ring' search interest (Figures 1&2).

The central argument running throughout is that a commercial significant gap exists in the UK smart wearable market - no product currently addresses the health tracking needs of the 10–18 age group through a ring form. Hence, that RYH is positioned to occupy that space before any established competitor can enter it (Kotler and Armstrong, 2018).

Market Selection

The UK smart wearables market is experiencing a rapid growth in recent years. According to Mordor Intelligence (2025) market size is expected to grow from USD 5.89 billion in 2025 to USD 6.81 billion in 2026 and is forecast to reach USD 14.02 billion by 2031 at 15.55% CAGR over 2026-2031. Ongoing NHS improvements, expansion of high-speed internet coverage and fintech-linked health-insurance incentives create a favorable environment for accelerating the demand for smart wearables. Mordor Intelligence (2025) identifies smart rings as one of the fastest-expanding segments, driven by consumer demand for discreet, screen-free health monitoring that does not replicate the notification-heavy experience of smartwatches. This distinction is commercially significant: Mintel (2026) reports that 43% of UK smartwatch users feel overwhelmed by excessive notifications.

Specifically, there is a significant increase in interest in smart rings worldwide and in the UK in 5 last years.

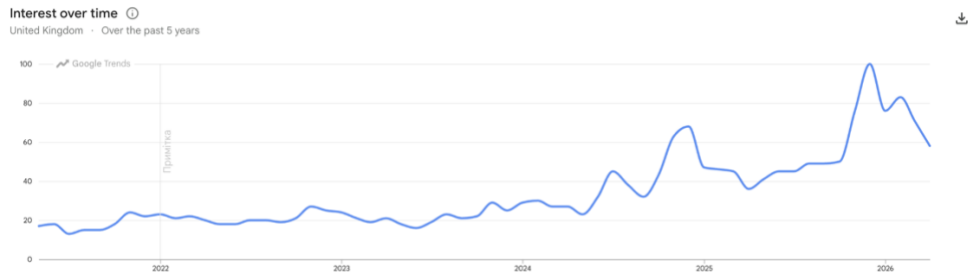


Figure 1: UK Google Trends search interest for “smart ring”, past five years. Source: Google Trend (2026a).

It presents the UK-specific search trajectory for 'smart ring'. The UK data reveals a three-stage growth pattern: a low-baseline period, a first acceleration phase coinciding with Samsung Galaxy Ring's UK launch and Oura Ring 4's expansion, and a peak of 100 in early 2026. Critically, search interest is settling at approximately 60–80 indicating mainstream awareness has been permanently established, reducing the consumer education burden typically associated with new launches (Chaffey and Ellis-Chadwick, 2022). Worldwide interest follows a similar pattern.

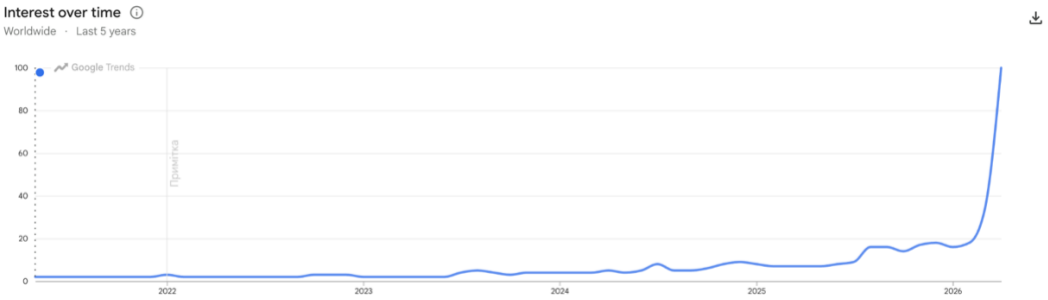


Figure 2: Worldwide Google Trends search interest for “smart ring”, past five years. Source: Google Trends (2026b).

Analysis of top UK related queries over five years (Appendix A1) shows Oura-branded terms each grew +450%, confirming Oura as the brand through which UK consumers know about the category. The +60% growth in 'best smart ring' confirms the market has moved from awareness to active product comparison over five years (Chaffey and Ellis-Chadwick, 2022). Most strategically significant is that no search query in the top 10 mentions children, teenagers, families, or junior health tracking. It confirms that the children's smart ring sub-category represents a structural gap in UK consumer search behaviour.

To evaluate the commercial potential of RYH, it is essential to estimate the size of the market opportunity. This approach allows us to move from a broad theoretical market

toward a realistic estimate of users that the product can capture. Because this project is focused on the UK market, the United Kingdom is used as the primary geographic scope for analysis.

Product Line	Market Tier	Size	Rationale & Sources
Kids Ring	TAM	6,510,000	All UK teenagers aged 10–18. Source: ONS Mid-Year Population Estimates (2024a).
	SAM	3,715,000 (57% of TAM)	Filtered by: smartphone ownership among 12–15s (96%) and 16–18s (98%) (Ofcom, 2024); and parent wearable purchase intent for children (IMARC, 2024).
	SOM	**186,000 (5% of SAM)**	Benchmarked against comparable DTC wearable new-entrant capture rates: Oura ~5% Year 1; Circular Ring ~3% Year 1; Whoop 4–6% Years 1–2 (Grand View Research, 2025).
Parent Ring	TAM	14,080,000	All UK adults aged 30–45. Source: ONS Mid-Year Population Estimates (2024a).
	SAM	2,790,000 (19.8% of TAM)	Filtered by: parents of a child aged 10–18 in 2026 (ONS Birth Statistics, 2024b: mean maternal age 31.0 years; mean paternal age 33.9 years); and adult wearable ownership/intent (IMARC, 2024).

Product Line	Market Tier	Size	Rationale & Sources
	SOM	**126,000 (4.5% of SAM)**	Secondary purchase driven by kids ring halo effect; benchmarked against adult DTC wearable new-entrant rates (Grand View Research, 2025).
Combined Year 3 SOM	**SOM (Yr 3)**	**312,000**	Total realistic user base by end of Year 3, assuming effective UK-wide DTC marketing, DTC distribution channel, and hardware production capacity.

Table 1: TAM–SAM–SOM Market Sizing Framework for RYH - Kids Ring and Parent Ring, United Kingdom. *Source: Primary analysis based on ONS (2024a, 2024b), Ofcom (2024), IMARC (2024), and Grand View Research (2025).*

The TAM-SAM-SOM framework illustrates realistic numbers of what we could obtain in the first years after launching and proving that there is a real potential in our selected market. However, this model relies only on secondary data assumptions. The purchase intent filter is based on analogous wearable categories (IMARC, 2024) rather than direct surveys. This is a genuine constraint on the precision of the SOM estimate.

STP-analysis

Segmentation:

Applying Kotler and Armstrong's (2018) segmentation framework, the RYH market is divided across three primary variables: demographic, psychographic, and behavioural.

Demographically, two distinct groups are identified. The first is UK teenagers aged 10–18, representing a total population of 6.51 million (ONS, 2024a). Further it was divided into two functional sub-groups: children aged 10-14 (Junior Mode users) and teenagers aged 14–18 (Teen Mode users). The age boundary is not arbitrary - at 13, UK GDPR grants minors independent data rights, making it a legal as well as a product design threshold. The second demographic is parents aged 30-45, estimated at 14.08 million in the UK (ONS, 2024a). The meaningful proportion of them have a child in the 10–18

bracket based on mean maternal and paternal age data (ONS, 2024b). Within this group, Ofcom (2024) reports that 96% of 12–15-year-olds and 98% of 16–18-year-olds own a smartphone. It confirms digital readiness across the target age range.

Psychographically, three user personas: Mia, Vladislav, and Oleg (Appendix B2-B4) were developed through secondary research and analysis of UK consumer behaviour in the health technology market. These personas were tested through primary qualitative interviews. The interviews therefore functioned as a stress-test of the product concept, examining the pain points and identifying the ideas for RYH's positioning as a basis for strategic decisions (Appendix B5). Each represents a distinct combination of lifestyle values, technology attitudes, and tolerance for parental oversight.

Behaviourally, the Jobs-to-be-Done framework (Christensen et al., 2016) reveals that each segment hires RYH for fundamentally different purposes. Functional jobs vary from sleep pattern understanding to exam-season support, emotional jobs include feeling in control of personal data, social jobs such as stylish appearance and not being monitored. This behavioural layer is critical because it explains why a single product experience cannot serve both the buyer (parent) and the user (teen) simultaneously. The survey sample (n=40; Appendix B1) provides partial validation but convenience sampling remains a limitation (Saunders et al., 2019).

Targeting:

RYH adopts a differentiated strategy, which provides distinct experiences for separate but interconnected segments (Kotler and Armstrong, 2018). The first primal target segment is 14-18 teens as users and the second is 30-45 years old parents – users and decision-makers for themselves and their children. The secondary target is children aged 10-14 since the final decision for their purchases are making their parents. This age boundary is reinforced by UK GDPR, which requires parental consent for processing the personal data of under-13s, which making the parent not only the commercial gatekeeper but the legal one.

Our primary research showed that parents and teens assigned different priorities: parents ranked data accuracy, privacy controls, and weekly summaries highest; teenagers ranked design customisation, peer sharing, and ease of use. These divergent priorities require Junior and Teen Modes to operate simultaneously but independently.

Positioning:

Competitor SWOT analysis (Appendix B6) reveals a consistent structural pattern across the five primary smart ring competitors: every adult ring brand such as Oura, Ultrahuman, Samsung, RingConn scores zero on family and community architecture. Also every children's device competitor such as Xplora, Fitbit Ace LTE, Garmin Bounce

2 operates in the smartwatch or band form factor and targets children below the age of 12. This pattern is visualised in two perceptual maps (Appendix B7).

The first map, plotted on axes of price versus community and family focus, shows all adult ring competitors clustered at near-zero on the Y-axis, leaving the upper-right quadrant entirely unoccupied. The second map, plotted on price versus customisation within the children's wearables market specifically, is more analytically significant for RYH's positioning. Fitbit Ace LTE is the only competitor with meaningful customisation, but targets ages 7–12 with a mandatory subscription and band form factor. Garmin Bounce 2 is the most expensive device in the category yet scores near-zero on customization. RYH Junior, at £149 with no mandatory subscription and the highest customisation score in the category through IP-collaboration ring faces, occupies the upper-right quadrant alone.

RYH positioning statement is “For families with children and teenagers aged 10-18 who want to track their health together – without anxiety and without surveillance. RYH (Ring your health) is a smart ring and family health platform that, unlike Oura and Galaxy Ring (adult versions only) and Fitbit Ace LTE (a kids fitness band for under-12s), delivers the first smart ring designed for growing bodies: paediatric-calibrated health sensors, a peer challenge community, IP-collaboration ring designs, and a parent dashboard with AI-generated plain-English health summaries because teenagers stop devices that feel like monitoring, but continue wearing devices they are proud to own.”

This statement scored 8.75 out of 10 in group evaluation (Clarity 9/10, Differentiation 9/10, Credibility 8/10, Defensibility 9/10). Credibility scored lowest because claims such as 'paediatric-calibrated sensors' and 'peer challenge community' imply capabilities that must be demonstrated at launch. It was identified in blind spot analysis (Seminar 3).

Applying Porter's (1980) competitive strategy framework, RYH is positioned as a focus differentiator. RYH is not competing on price with budget rings (RingConn, Amazfit) nor on ecosystem breadth with Samsung or Apple, but on deep specialisation in an entirely unserved niche.

Three structural competitive moats sustain this position. First, first-mover category ownership. Second, a proprietary teen biometric dataset that trains the AI interpretation layer and compounds in value over time. Third, community and lifecycle network effects: the size-up programme creates an eight-year brand relationship from age 10 to adult graduation.

Product launch strategy

This section applies Kotler and Armstrong's (2018) Marketing Mix framework to RYH's UK launch, addressing product, price, place, and promotion. It is worth noting that while the traditional 4Ps remain a structurally sound analytical framework, Sulaj and Pfoertsch (2024) argue that digital-first brands require an expanded model that accounts for

experience, exchange, and evangelism - dimensions that are embedded throughout this section rather than treated as separate elements.

Product Strategy

RYH launches two hardware products: the RYH Junior ring (£149, ages 10–18) and the RYH Adult ring (£199, ages 18+). Both pair with an AI-powered companion app but serve different purposes within the product family. The Junior ring is the primary acquisition product - the first ring a child receives. The Adult ring is the graduation product, designed to receive teenagers who complete the size-up programme at approximately 17–18 years old and transition to a full adult health suite.

The Junior ring tracks sleep quality, heart rate variability (HRV), blood oxygen saturation (SpO_2), resting heart rate, skin temperature, and physical activity, with all metrics calibrated for growing bodies. Two experience modes operate simultaneously within a single family account: Junior Mode (ages 10–14), in which parents have full data visibility and approve community participation; and Teen Mode (ages 14–18), in which teenagers control their own sharing level - Full, Summary Only, or Off and manage their own community connections independently.

The modular insert system is the product's most structurally important feature and requires precise description. The insert is a flexible inner liner that fits inside the ring shell. Its dual function is first, to accommodate finger growth between ring replacements. The insert allows the same ring shell to be worn comfortably for approximately 12–18 months. Secondly, it enables design customisation without purchasing a new ring. Inserts are available in different colourways and limited-edition designs. It is directly addressing the social JTBD identified in primary research.

Beyond inserts, IP-collaboration ring designs are available as activity-unlocked exclusives and seasonal drops. These are limited-edition ring faces and inserts released in partnership with film, animation, or cultural IP brands.

The product also includes a community space. This community is non-competitive: progress is shown as percentage improvement against each user's own personal results, not as a ranking against others. This decision directly addresses the community risk identified in primary research (interviews) and rankings were rejected by teen participants as anxiety-inducing and socially pressuring. Survey evidence (Appendix C2) ranked activity tracking, daily wellbeing explanations, and exam-season support as top features.

Pricing Strategy

RYH's pricing architecture is built around Nagle and Müller's (2018) principle of value-based pricing: price anchored to the perceived value delivered to the buyer rather than the cost of production. Primary research interviews confirmed that £120 represents the

price anchor at which parents feel comfortable, £150 is the ceiling of acceptability, and £200 or above triggers rejection.

Category	Product / Tier	Price	Key Inclusions
HARDWARE			
Kids & Teen Ring (Ages 10–18)	**RYH Junior**	**£149 one-off**	Modular insert kit (sizing + design customisation) Dual Junior / Teen experience modes IP-collaboration custom designs and seasonal drops Activity-unlocked exclusive ring faces 3 months RYH Connect included free
Adult Ring (Ages 18+)	**RYH Adult**	**£199 one-off**	Full adult health suite: HRV, SpO_2, recovery AI plain-English personal health journal Family account management Premium design finishes Health history continuity from Junior ring 3 months RYH Connect included free
SIZE-UP PROGRAMME			
Stage 1 - Starter (Ages 10–11)	RYH Junior	**£149**	First ring; full modular insert kit included. Baseline price.
Stage 2 - Explorer (Ages 12–14)	RYH Junior	**£109 (save £40)**	Replacement ring at discounted loyalty price. Saving £40 vs full price.

Category	Product / Tier	Price	Key Inclusions
Stage 3 - Senior (Ages 15–17)	RYH Junior	**£89 (save £60)**	Replacement ring at further discounted loyalty price. Saving £60 vs full price.
Stage 4 - Graduate (Ages 17–18+)	RYH Adult	**£149 (£50 loyalty discount applied)**	Graduation to adult ring at £50 loyalty discount vs. standard £199 adult price. Health history carries forward seamlessly.
Full Journey Total	4 rings	**£496**	Saving £100+ vs buying a new ring at full price at each stage (£596+). Journey validated against primary research price sensitivity data (Seminar 4, 2026).
SUBSCRIPTIONS -ALWAYS OPTIONAL			
Free Tier	**RYH Basic**	**£0 forever**	Steps, sleep, resting HR, SpO_2 7-day data history Daily emoji health card (child) Basic parent view - activity only Activity milestone badges No AI summaries, no community, no full parent dashboard Ring health tracking never stops without a subscription
Personal Tier	**RYH Connect**	**£3.99/mo or £39.99/yr (save £7.89)**	Everything in Basic + unlimited data history Full metrics: HRV, sleep stages, recovery Weekly AI summary -parent plain English

Category	Product / Tier	Price	Key Inclusions
			Weekly AI summary -teen version Pattern alerts + illness early warning Traffic-light health status Community space + friends + school groups Full ring face design library + 48h early access GP-ready health data export
Family Tier	**RYH Family**	**£6.99/mo or £69.99/yr (save £13.89) Up to 3 rings**	Everything in Connect (all rings) Unified family dashboard -all rings Family AI weekly report Custom family challenges + group chat Location check-in (opt-in, child-visible always) Priority alerts -push and email 1 free annual insert set per child ring 7-day first access to design drops Priority support -24h response

Table 2: RYH Full Pricing Architecture - Hardware, Subscription Tiers, and Size-Up Programme. Source: Group research (Seminar 6); validated against interview price sensitivity data (Seminar 4)

Survey evidence validates the £149 price point (Appendix C2): 89% of teens accepted £150–£199, with 0% accepting £220+. It places the launch price within the comfort range with headroom for IP-collaboration premiums.

Both hardware products include **3 months of RYH Connect free**, reducing the initial adoption barrier. RevenueCat (2025) reports that freemium apps achieve a median conversion rate of 2.18% from free to paid, while Health and Fitness apps reach 12.1% at the 90th percentile under hard paywall models suggesting that RYH's opt-in trial strategy is well-calibrated to convert a meaningful proportion of hardware buyers into ongoing subscribers.

Place Strategy

RYH launches as a **direct-to-consumer (DTC) digital-first brand**, the RYH website as a single primary sales channel. There is no retail distribution at launch. Mu and Yi (2024) demonstrate that DTC models create significantly higher consumer loyalty than retail intermediaries. For RYH this matters because the product requires detailed onboarding impossible through third-party checkout; the family relationship sustaining subscription revenue is built through brand communication, not marketplace reviews; and direct sales preserve the margin funding the AI layer and community.

Microsoft Clarity testing (Appendix C3) validated the architecture: 21 sessions, 19 unique users, 46.33% scroll depth, 23s active time; 90.5% new visitors; 19.05% dead clicks flagging pre-launch UX friction. Click heatmap (Appendix C4) shows the four most-clicked elements were evenly distributed - 'For parents', 'Pricing', the waitlist input, and 'For teens' each at 7.5%, it confirms the dual-audience navigation functions as designed. 90% of survey respondents found the website easy or very easy to navigate (Appendix C5). The secondary channel is the App Store and Google Play. School partnerships are flagged (Seminar 1) as Phase 2 but excluded from launch given 12–18 month UK procurement timelines.

Promotion Strategy

RYH's promotion strategy is architecturally split between two audiences who require different approaches: parents as buyers, and teenagers as users. This split is not a stylistic preference - it is legally required. The UK CAP Code restricts advertising directed at children, making paid acquisition targeting the teen segment through conventional advertising channels impermissible regardless of commercial preference.

Parent-facing acquisition operates across three channels. Content marketing such as exam-season sleep guides, screen-time and recovery research, parent conversation guides - serves dual purposes as both SEO traffic generation and trust-building with the parent segment. All content points back to the RYH website as the primary destination. Paid social advertising on Facebook and Instagram, targeted at the 30–45 demographic, leads with the parent value proposition confirmed in primary research: "Understand how your child is doing without making them feel watched."

Teen-facing retention relies on the product. The social JTBD analysis established that teenagers sustain engagement with products they are proud to wear publicly; a ring worn at school generates peer curiosity, and custom design makes it a conversation object. TikTok and Instagram Reels - customisation, drop reveals, goal achievement -reinforce this through organic community seeding (Kaplan and Haenlein, 2010).

The CAP Code constraint means teen-facing brand awareness depends entirely on word-of-mouth, making wear compliance and hardware quality the foundation of the entire promotional strategy.

Ethical Marketing

RYH collects continuous biometric and health data from children aged 10–18. Under UK GDPR Article 9, health and biometric data constitute **Special Category Data**, for which processing is prohibited unless one of ten conditions is met (ICO, 2023). The applicable condition for RYH is explicit consent - more stringent than standard consent for ordinary personal data.

Primary research reinforces commercial significance (Appendix C8): when parents were asked what would make them hesitate before buying RYH, data privacy was the single largest concern, exceeding price, subscription cost, accuracy, and durability.

The Children's Code is a statutory code under the Data Protection Act 2018 setting out 15 standards designed to ensure that the best interests of the child are the primary consideration in service design (ICO, 2020). Standard 14 explicitly cites fitness bands recording activity, placing RYH unambiguously within scope. Five standards shape RYH's architecture: Best Interests (Standard 1) anchors decisions where commercial interest and child wellbeing conflict; DPIA (Standard 2) is a pre-launch legal requirement; Default Settings (Standard 7) require high privacy by default - Teen Mode's Summary Only default is a legal compliance mechanism, not a product feature; Data Minimisation (Standard 8) is satisfied by the Basic tier; Nudge Techniques (Standard 13) are prohibited, structurally preventing anxiety-inducing upgrade prompts.

The geolocation is removed from the list of our features. Although Standard 10 permits opt-in geolocation, primary research found that teenagers consistently rejected location tracking even with parental controls.

Under the Data Protection Act 2018, the age of digital consent in the UK is 13 (ICO, 2023). Under-13s require verifiable explicit parental consent before any data processing begins. Children aged 13 and over can consent independently, though because RYH processes Special Category health data, that consent must be explicit. The onboarding flow must therefore include a distinct, age-appropriate consent step for health and biometric data. The 13-year boundary is also the legal trigger for the automatic Junior-to-Teen Mode transition.

CAP Code Rule 5.4.2, updated to reflect the Digital Markets, Competition and Consumers Act 2024 (in force April 2025), prohibits marketing communications from making direct appeals to children to buy a product or to persuade their parents to buy one for them (ASA/CAP, 2025). This enforces the promotional split in Section 4.

The JMIR scoping review of mobile and wearable AI in child and adolescent psychiatry identifies the lack of randomised controlled trials as the largest gap in the field, with most existing studies being small pilot trials focused on diagnosed clinical conditions such as ASD and ADHD (Dwyer et al., 2022). Three implications follow: RYH must not make clinical efficacy claims in its marketing; the AI interpretation layer must not present insights as medical conclusions; and an independent clinical ethics review by a child psychologist before launch.

Critical reflection

A fundamental constraint on the secondary research underpinning this plan is that the children's smart ring market does not yet exist. No competitor product serves the 10–18 age group through a ring form factor, which means no market reports, consumer surveys, or industry analyses cover this specific commercial environment. The TAM–SAM–SOM model is the most visible expression of this limitation. As Kumar and Reinartz (2018) observe, market sizing models in pre-launch categories are projections built on assumptions; their precision is bounded by the quality of the analogue data, not by the calculation method.

Primary research presented an even greater practical challenge. As university students, the group does not routinely interact with the two target segments - children aged 10–18 and parents aged 30–45. Recruitment was therefore conducted through personal networks: interview invitations and Google Form survey links were distributed through group chats and to acquaintances, friends, and relatives. While this approach successfully reached participants from both segments, it produced a structural weakness. The resulting sample is non-random and reliant on convenience and snowball sampling - methods that introduce selection bias and limit the generalisability of findings to the wider UK population (Saunders et al., 2019).

This plan addresses market entry, segmentation, positioning, distribution, promotion, and ethical compliance, but it does not address the hardware side of the product. The cost of paediatric-calibrated sensor manufacturing, supply chain configuration, mechanical design for children's finger sizes, and the data infrastructure required to train the AI interpretation layer are all outside the scope of a marketing plan but inside the scope of whether the product can actually exist as described. The blind spot analysis (Group research, Seminar 3, 2026) flagged the absence of a hardware strategy.

The limitations above do not undermine the value of the data-driven approach taken throughout this plan. At every stage such as market selection, segmentation, targeting, positioning, and launch strategy - strategic decisions were anchored to evidence rather than intuition. Google Trends data established that the smart ring category has reached mainstream UK awareness. Primary research challenged assumptions about teen privacy expectations, parent oversight preferences, and price sensitivity that the group had initially treated as obvious. The location check-in feature was removed from the Family tier (Ethical Marketing Section) specifically because survey evidence contradicted the group's initial design intuition. It is a meaningful illustration of how data-driven marketing reduces the risk of decisions taken on intuition alone (Chaffey and Ellis-Chadwick, 2022).

The plan was developed using a combination of analytical tools: Google Trends for category demand analysis, Google Forms for primary research data collection, Google Analytics and Microsoft Clarity for website prototype usability data, and Anthropic's Claude and OpenAI's ChatGPT as research assistants for source identification. AI tools were used throughout the research under the strict supervision.

A final theme cuts across the plan: the tension between data and ethics. RYH's long-term advantage rests on proprietary l biometric data; the same data is Special Category under UK GDPR. More data improves the AI; more data also increases regulatory and reputational exposure. This is a tension to be managed continuously, not a problem to be solved.

Conclusion

This plan has argued that a commercially significant and ethically navigable gap in the UK smart wearable market: no product currently serves the 10–18 age group through a ring form factor, despite five years of structural growth in category demand (Google Trends, 2026) and a UK market projected to reach $26.51 billion by 2033 (IMARC, 2024). RYH is positioned to occupy that gap as a focus differentiator (Porter, 1980), targeting parents as buyers and teenagers as users through a differentiated product architecture, a direct-to-consumer launch, and a freemium subscription model anchored at £149 for the Junior ring and £199 for the Adult ring.

RYH's competitive advantage is therefore not located in any single feature, price point, or marketing channel - it lies in the integrity of the whole system. Being the first brand to establish trusted, evidence-informed health tracking for children in the UK is a position that grows stronger the longer it is held, and one that competitors cannot replicate by simply launching a similar product. The opportunity is to define what responsible health technology for young people looks like and to make that definition the standard others have to meet.

Reference list:

ASA/CAP (2025) *Children: Promotional marketing -Rule 5.4.2*. London: Advertising Standards Authority / Committee of Advertising Practice. Available at: https://www.asa.org.uk/advice-online/children-promotional-marketing.html (Accessed: May 2026).

Boer, A., Batoux, C., Pechoux, C., Bergeret, L., Cotta-Lacombe, A. and Duclos, M. (2023) 'Effects of commercially available wearable devices on physical activity promotion and health in children and adolescents: systematic review', *Applied Sciences*, 13(12), p. 7194. Available at: https://www.mdpi.com/2076-3417/13/12/7194 (Accessed: May 2026).

Chaffey, D. and Ellis-Chadwick, F. (2022) *Digital marketing: strategy, implementation and practice*. 7th edn. Harlow: Pearson.

Christensen, C.M., Hall, T., Dillon, K. and Duncan, D.S. (2016) 'Know your customers' "jobs to be done"', *Harvard Business Review*, 94(9), pp. 54–62. Available at: https://hbr.org/2016/09/know-your-customers-jobs-to-be-done (Accessed: May 2026).

Damme, K.S.F., Tozzi, L., Cheng, D., Barbey, A.K., Letkiewicz, A.M. and Mittal, V.A. (2024) 'Physical and mental health in adolescence: novel insights from a transdiagnostic examination of FitBit data in the ABCD study', *npj Mental Health Research*. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC10837202/ (Accessed: May 2026).

Dwyer, M.J., Pasini, M., De Dominicis, S. and Righi, E. (2022) 'Use of mobile and wearable artificial intelligence in child and adolescent psychiatry: scoping review', *Journal of Medical Internet Research*, 24(3), e33560. Available at: https://www.jmir.org/2022/3/e33560 (Accessed: May 2026).

Elgendi, M., Markov, K., Liu, H., De Vos, M., Khalaf, K., Khandoker, A., Jelinek, H.F., Goswami, D. and Menon, C. (2026) 'Wearable devices for anxiety assessment: a systematic review', *Communications Medicine*. Available at: https://www.nature.com/articles/s43856-025-01234-6 (Accessed: May 2026).

Google Trends (2026a) *Interest over time: 'smart ring', Worldwide, January 2021–May 2026*. Available at: https://trends.google.com (Accessed: May 2026).

Google Trends (2026b) *Interest over time: ‘smart ring’, United Kingdom, January 2021–May 2026*. Available at: https://trends.google.com (Accessed: May 2026).

Google Trends (2026c) *Popular related queries: ‘smart ring’, United Kingdom, last 5 years*. Available at: https://trends.google.com (Accessed: May 2026).

Grand View Research (2025) *Smart rings market size, share & trends analysis report 2025–2033*. Available at: https://www.gminsights.com/industry-analysis/smart-rings-market (Accessed: May 2026).

Group research (2026a) *Seminar 1: Market analysis and Google Trends data*. Unpublished seminar work. Module 32238 ‘Marketing and New Technology’, University of Birmingham.

Group research (2026b) *Seminar 2: Persona development and customer segmentation*. Unpublished seminar work. Module 32238 ‘Marketing and New Technology’, University of Birmingham.

Group research (2026c) *Seminar 3: Competitive analysis, SWOT matrices, perceptual maps and blind spot analysis*. Unpublished seminar work. Module 32238 ‘Marketing and New Technology’, University of Birmingham.

Group research (2026d) *Seminar 4: Primary qualitative interviews, Google Form survey and prototype website testing*. Unpublished seminar work. Module 32238 ‘Marketing and New Technology’, University of Birmingham.

ICO (2020) *Age appropriate design: a code of practice for online services*. Wilmslow: Information Commissioner’s Office. Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/childrens-code-guidance-and-resources/age-appropriate-design-a-code-of-practice-for-online-services/ (Accessed: May 2026).

ICO (2023a) *What is special category data?* Wilmslow: Information Commissioner’s Office. Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/special-category-data/what-is-special-category-data/ (Accessed: May 2026).

ICO (2023b) *Using children’s information: a guide*. Wilmslow: Information Commissioner’s Office. Available at: https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/using-childrens-information-a-guide/ (Accessed: May 2026).

IMARC Group (2024) *UK smart wearables market: size, share & industry analysis 2025–2033*. Available at: https://www.imarcgroup.com/uk-smart-wearables-market (Accessed: May 2026).

Kaplan, A.M. and Haenlein, M. (2010) ‘Users of the world, unite! The challenges and opportunities of social media’, *Business Horizons*, 53(1), pp. 59–68. doi: 10.1016/j.bushor.2009.09.003.

Kotler, P. and Armstrong, G. (2018) *Principles of marketing*. 17th edn. Harlow: Pearson.

Kumar, V. and Reinartz, W. (2018) *Customer value management: strategies for building profitable customer relationships*. 3rd edn. Berlin: Springer.

Lutz, L., Becker, S., Bhatt, S., Stadelmann, S., Dapp, L. and Kohls, N. (2025) ‘Associations between wearables vital parameters and self-perceived mood: an ecological momentary assessment study among healthy adolescents’, *Frontiers in Psychology*. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC12777082/ (Accessed: May 2026).

Mintel (2026) *UK smartwatches and wearable technology market report*. London: Mintel. Available at: https://store.mintel.com/report/uk-smartwatches-and-wearable-technology-market-report (Accessed: May 2026).

Mordor Intelligence (2025) *United Kingdom wearable technology market -size, share and forecast to 2030*. Available at: https://www.mordorintelligence.com/industry-reports/united-kingdom-wearable-technology-market (Accessed: May 2026).

Mu, W. and Yi, Y. (2024) ‘The impact of characteristic factors of the direct-to-consumer marketing model on consumer loyalty in the digital intermediary era’, *Frontiers in Psychology*, 15, 1347588. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC10944917/ (Accessed: May 2026).

Nagle, T.T. and Müller, G. (2018) *The strategy and tactics of pricing: a guide to growing more profitably*. 6th edn. Abingdon: Routledge.

Ofcom (2024) *Children and parents: media use and attitudes report 2024*. London: Ofcom. Available at: https://www.ofcom.org.uk/media-use-and-attitudes/media-habits-children/children-and-parents-media-use-and-attitudes-report-2024 (Accessed: May 2026).

ONS (2024a) *Annual mid-year population estimates: mid-2024*. Newport: Office for National Statistics. Available at: https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/bulletins/annualmidyearpopulationestimates/mid2024 (Accessed: May 2026).

ONS (2024b) *Births in England and Wales: summary tables*. Newport: Office for National Statistics. Available at: https://www.ons.gov.uk/peoplepopulationandcommunity/birthsdeathsandmarriages/livebirths (Accessed: May 2026).

Porter, M.E. (1980) *Competitive strategy: techniques for analyzing industries and competitors*. New York: Free Press.

RevenueCat (2025) *State of subscription apps 2025*. Available at: https://www.revenuecat.com/state-of-subscription-apps-2025/ (Accessed: May 2026).

RYH (2026) *RYH -Ring Your Health* [website]. Available at: https://dzaagrebiina.github.io/RYH/#home (Accessed: May 2026).

Saunders, M., Lewis, P. and Thornhill, A. (2019) *Research methods for business students*. 8th edn. Harlow: Pearson.

Sulaj, K. and Pfoertsch, W. (2024) 'Introducing 5Es marketing-mix: a new framework for effective marketing in the digital age', *Academy of Strategic Management Journal*, 23(2). Available at: https://www.abacademies.org/articles/introducing-5es-marketingmix--a-new-framework-for-effective-marketing-in-the-digital-age.pdf (Accessed: May 2026).

Thompson, L., Baragwanath, G., King, B., Sheringham, J. and Mays, N. (2025) 'Exploring the use of smartwatches and activity trackers for health-related purposes for children aged 5 to 11 years: systematic review', *Journal of Medical Internet Research*, 27, e62944. Available at: https://www.jmir.org/2025/1/e62944 (Accessed: May 2026).

UN (2015) *Transforming our world: the 2030 agenda for sustainable development*. New York: United Nations. Available at: https://sdgs.un.org/2030agenda (Accessed: May 2026).

Zhang, Y., Bhatt, D., Bhatt, P., Solomon, A., Bhattacharya, A., Bhattacharya, S. and Ghosh, S. (2025) ‘Using a consumer wearable activity monitoring device to study physical activity and sleep among adolescents: cohort study’, *JMIR mHealth and uHealth*, 13, e59159. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC11813160/ (Accessed: May 2026).

Appendices

Appendix A - Market Selection

Google Trends -Popular Related Queries (United Kingdom)

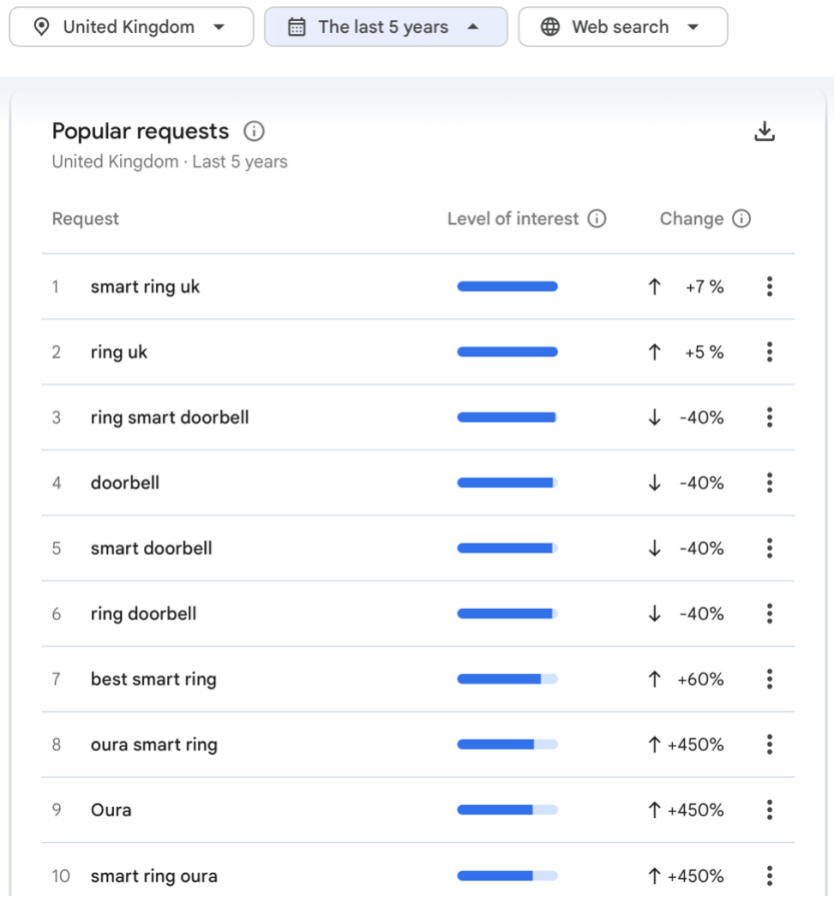


Figure 1: Google Trends -Top related queries for 'smart ring', United Kingdom, last 5 years. Oura-branded terms each grew by +450% across the period, confirming Oura as the brand through which UK consumers learned the category. 'Best smart ring' grew +60%, indicating rising consumer sophistication. The geographic qualifier 'smart ring uk' at position 1 (+7%) confirms purchase-intent searching is a structural and growing behaviour. Note: positions 3–6 relate to Amazon's Ring doorbell brand and are excluded from analysis. Most strategically significant -no term across the top 10 contains any reference to children, teenagers, families, or junior health tracking. Source: Google Trends (2026), accessed May 2026

Appendix B -STP Analysis

Supporting figures and tables for Section 3 of the main report.

B1. Survey Respondent Distribution

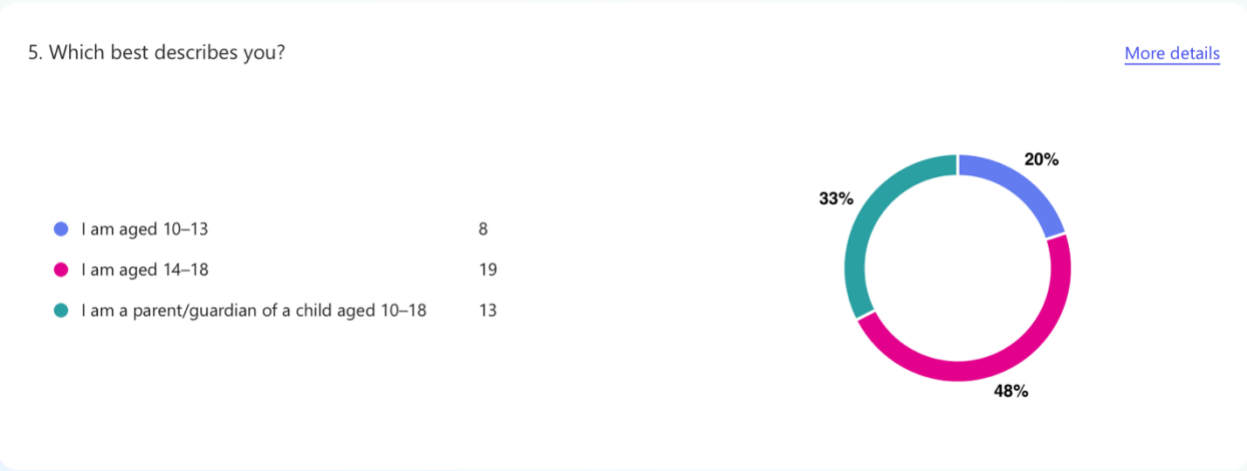


Figure 2: Survey respondent distribution by segment (Google Form Q5, n=40). 48% teenagers aged 14–18, 32% parents, 20% children aged 10–13 -closely matching the target audience distribution the plan is designed to serve. Source: Primary research (Group research, Seminar 4, 2026)

B2 User Persona - Mia

Mia, 39 – The Health-Conscious Protective Parent

Attribute	Details
Demographics	Age 35–45, lives in the UK, middle-income household, parent of a 13-year-old daughter, works full-time, responsible for family health decisions
Psychographics	Caring, protective, health-aware, wants to support her child without being too controlling. Interested in technology that helps her understand her child's wellbeing, but cautious about privacy and safety
Behaviours	Regularly checks her daughter's school routine, sleep habits and stress levels. Buys products that feel useful, safe and good value. Prefers simple explanations instead of complicated health data
Frustrations	Does not always know if her child is tired, stressed or not sleeping properly. Finds adult wearable apps too technical. Worries that her daughter may not openly talk about sleep, stress or health problems
Key Need	"I want to understand how my child is doing without making her feel watched."
Value We Offer	Parent-friendly wellbeing insights, simple health explanations, privacy-conscious family sharing and early awareness of sleep or routine problems

Source: Group research (Seminar 2); validated against primary interview data (Seminar 4)

B3. User persona - Vladislav

Vladislav, 16 – The Style-Conscious Teen Fitness Beginner

Attribute	Details
Demographics	Age 16, secondary school student in the UK, recently started going to the gym, limited personal budget, influenced by friends, social media, fitness content and fashion trends
Psychographics	Ambitious, image-conscious, socially motivated, wants to improve his sleep, fitness and general health. Likes technology that feels modern, stylish and not childish
Behaviours	Goes to the gym several times a week, tries to improve his routine, compares progress with friends, uses apps when they are simple and visually engaging. Interested in products that look fashionable and can be personalised
Frustrations	Cannot afford expensive smart rings from competitors. Finds many health apps too serious or confusing. Wants to track progress but does not want boring charts or advice that feels like a lecture
Key Need	“I want a cool smart ring that helps me improve my fitness and sleep without being too expensive.”
Value We Offer	Affordable smart ring, fashionable custom design, simple progress tracking, teen-friendly explanations and optional sharing with friends

Source: Group research (Seminar 2); validated against primary interview data (Seminar 4)

B4. User persona - Oleg

Oleg, 42 – The Sport-Focused Single Father

Attribute	Details
Demographics	Age 40–45, single father living in the UK, has a 12-year-old son, active lifestyle, interested in sport, works full-time and manages parenting responsibilities alone
Psychographics	Responsible, disciplined, family-oriented, believes sport builds confidence and healthy habits. Wants to encourage his son to be active, but does not want to pressure him too much
Behaviours	Encourages his son to join sports clubs, go outside and build healthier routines. Looks for products that support long-term

	development, motivation and family connection. Values durability, affordability and practical benefits
Frustrations	Finds it difficult to keep his son motivated consistently. Wants to support his son's health but does not always have time to monitor sleep, activity and routine. Worries that adult fitness trackers may not be suitable or engaging for a younger teenager
Key Need	"I want to help my son build healthy habits in a way that feels fun and motivating."
Value We Offer	Family-connected wellbeing tracking, sport-friendly motivation, simple progress updates, affordable pricing and ring replacement as the child grows

Source: Group research (Seminar 2); validated against primary interview data (Seminar 4)

B5. Job-to-be-done (JTBD framework)

Job Type	Job Description	Customer	Product / Marketing Implication
Functional	Understand sleep quality and patterns	Teens + Parents	Sleep tracking is the MVP centrepiece; show patterns, not just scores
Functional	Translate confusing health data into simple explanations	Teens + Parents	Insight cards: what happened, why, and one practical next step
Functional	Support exam-season routines	Mainly teens	Exam Mode with calm sleep, revision, and recovery guidance
Emotional	Feel understood, not judged	Teens	Calm, supportive language; avoid guilt, blame, or "you failed" framing

Job Type	Job Description	Customer	Product / Marketing Implication
Emotional	Feel in control of personal data	Teens	Privacy choices in onboarding; teen-controlled sharing and data deletion
Emotional	Feel reassured the product is worth the money	Parents	Show value clearly: warranty, durability, battery life, no forced subscription
Social	Look mature, stylish -not childish	Teens	Minimalist, matte black, silver, grey and subtle custom design options
Social	Avoid being seen as monitored by parents	Teens	Do not lead with parent-control benefits in teen-facing marketing
Social	Share achievements but not private health data	Teens	Opt-in sharing of streaks/goals only -no public stress or sleep rankings

Source: Primary research - qualitative interviews (Seminar 4); framework: Christensen et al. (2016)

B6. Competitors SWOT analysis

SWOT: Oura (Oura Ring + App)

Strengths	Weaknesses
Category leader, strong brand trust and recognition	Expensive subscription (£5.99/month) causes friction
Accurate sensors, strong sleep/recovery data	High price

Premium positioning attracts health-conscious consumers	App feels clinical and impersonal; not emotionally supportive
Opportunities	**Threats**
Growing smart ring market	New entrants (Ultrahuman, RingConn) offering no-subscription models
Expanding into corporate wellness programmes	Samsung Galaxy Ring entering market with massive ecosystem
Growing female health tracking features	Users shifting to cheaper alternatives during cost-of-living pressures

SWOT: Ultrahuman (Ring Air + App)

Strengths	**Weaknesses**
No subscription model – strong selling point vs Oura	Accuracy complaints reported (sleep/step discrepancies)
Performance + metabolic focus appeals to biohackers	Smaller brand recognition vs Oura in UK market
Competitive pricing for hardware	Users report “struggling to understand all the data and scores”
Opportunities	**Threats**
Growing “no subscription” consumer preference	Oura’s brand dominance makes market penetration harder
Expansion into metabolic health (CGM integration)	Accuracy issues could damage trust long-term

SWOT: Xplora (Watch + App)

Strengths	**Weaknesses**
Strong parent-focused positioning	Customer service and product quality concerns
Safety features	Less suitable for older teenagers
School Mode	Monthly tariff required
Opportunities	**Threats**
Positioning around school safety	Cheaper kids smartwatches

Growing demand for child-safe technology	Older children may prefer smartphones or more mature wearables

SWOT: Apple Health / Apple Watch

Strengths	**Weaknesses**
Massive default health platform – huge user base	Users describe it as “data rich and insight poor”
No subscription for health features	Sleep data inconsistency is a common complaint
Deep ecosystem integration (iPhone, iPad, Mac)	Generic wellness approach, not specialised for any segment
Strong battery life	App insight
Opportunities	**Threats**
Could acquire or partner with interpretation-layer startups	Privacy concerns with Big Tech handling health data
Apple Intelligence AI could add personalised summaries	Users seeking more specialised, niche wellness solutions

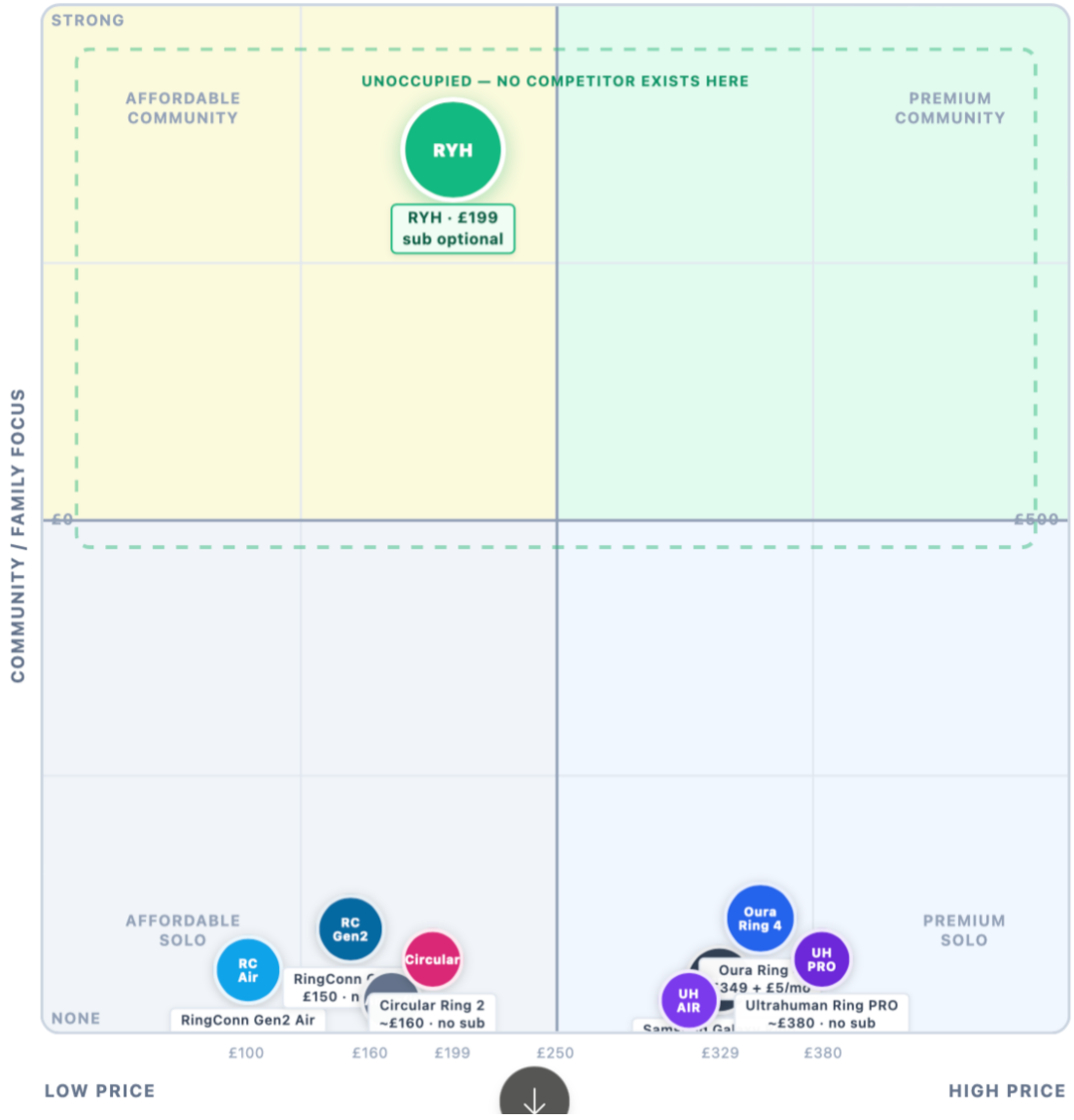
SWOT: RingConn (Ring + App)

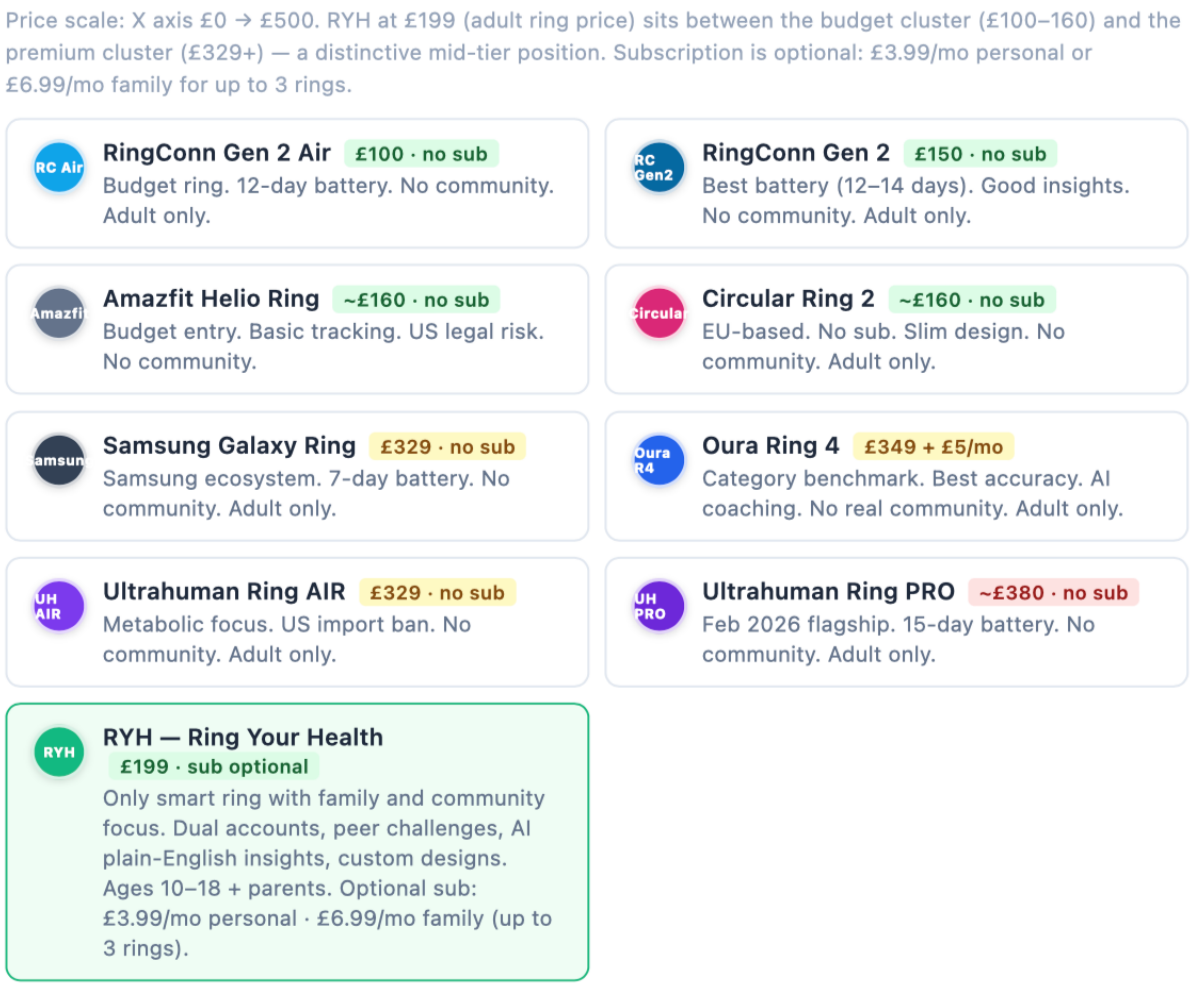
Strengths	**Weaknesses**
Budget/value positioning – most affordable smart ring	Limited insight depth compared to Oura
No subscription required	Smaller community and brand recognition
Simple interface appeals to casual users	Less accurate sensors vs premium competitors
Strong battery life	App insights may be less advanced than competitors such as Oura
Opportunities	**Threats**
Price-sensitive students and first-time ring buyers	Easily outcompeted if Oura/Samsung drop prices
Growing UK distribution through Amazon	Limited R&D budget vs well-funded competitors

Potential to expand into younger or teen markets because most smart rings are adult-focused	Strong premium competitors may dominate customers attention

B7. Positioning Matrix (perceptual Map)

7.1 - RYH positioning matrix for adult market





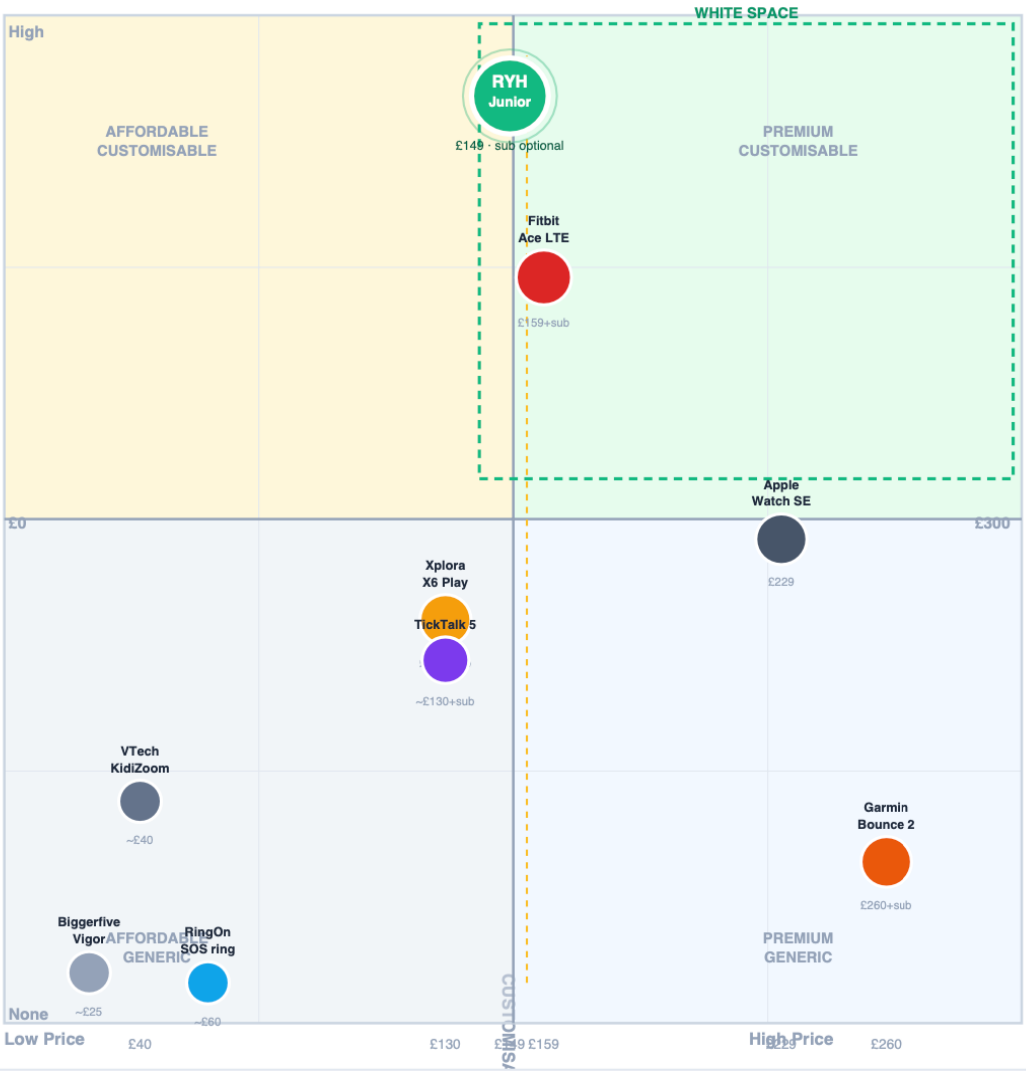
Explanation for RYH Positioning matrix for adult market

7.2 - RYH Positioning matrix for kids' market

RYH — Ring Your Health · Competitive Positioning

Kids Smart Devices Market

Matrix axes: X — Price (low to high) · Y — Customisation (none to high)



Biggerfive Vigor 2 ~£25 · no sub
Ultra-budget band. Step counting only. No customisation. Ages 4-16.

VTech KidiZoom DX3 ~£40 · no sub
Budget camera watch. Basic avatar games. Limited health tracking. Ages 4-12.

RingOn SOS ring ~£60 · pre-launch
Only other kids ring. Safety SOS only. No health tracking, no customisation. Ages 6-14.

Xplora X6 Play £130 + sub
4G smartwatch. Calling, GPS, Goplay gamification. Mid customisation. Ages 6-12.

TickTalk 5 ~£130 + sub
Smartphone-replacement watch. Video calling. Mid customisation. Ages 5-12.

Fitbit Ace LTE £159 + £2.99/mo mandatory
Best-in-class gamification. Customisable avatar earned via activity. Ages 7-12. Daily charging.

Apple Watch SE £229 · no sub
Premium teen option. Watch faces, Family Setup. Adult-designed. Ages 12+.

Garmin Bounce 2 £260 + £10/mo mandatory
Most expensive. Premium GPS, sports tracking. Very limited customisation. Ages 6-12.

RYH Junior £149 · sub optional
Only smart ring with health monitoring for 10-18s. Highest customisation: IP collabs, interchangeable designs, ring face personalisation. No mandatory subscription.

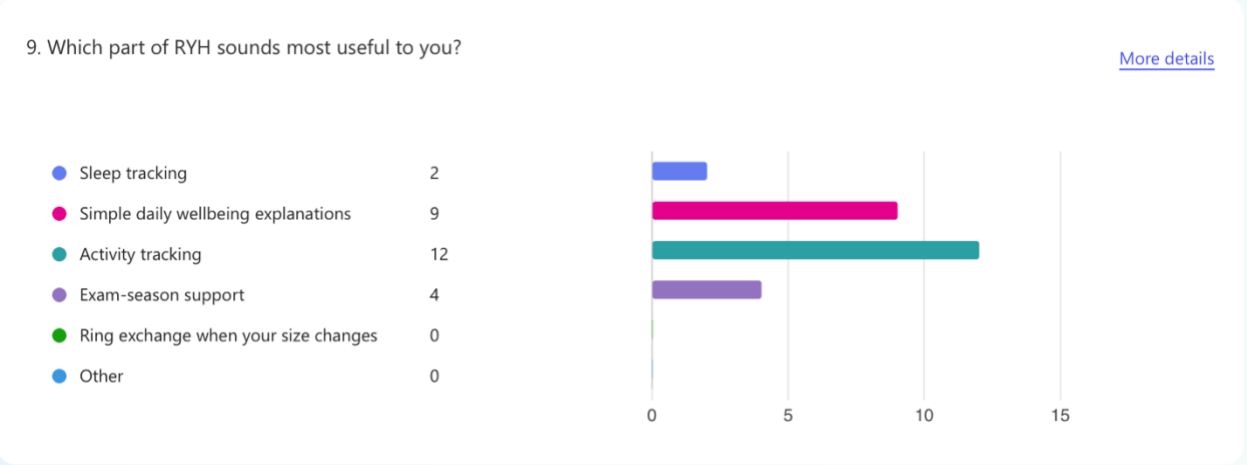
Key Insight

RYH Junior at £149 is the only smart ring in the kids device market. Fitbit Ace LTE is the only competitor with meaningful customisation (£159 + mandatory subscription) but targets ages 7-12 with a band form factor and requires daily charging. RYH sits at nearly the same price point but significantly higher on the Y-axis — offering the highest customisation in the category, a ring form factor, ages 10-18, and no mandatory subscription. The upper-right quadrant (premium and highly customisable) is the white space RYH owns.

Explanation for RYH positioning market for kids' market

Appendix C – Survey and web analytics results

C1 – Most useful features



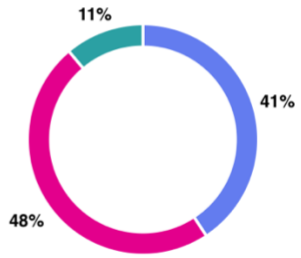
Google Form Survey results (n=40, Q=36)

C2 – Price sensitivity

20. What price feels reasonable for the RYH teen smart ring?

More details

£150–£179	11
£180–£199	13
£200-£220	3
£220+	0

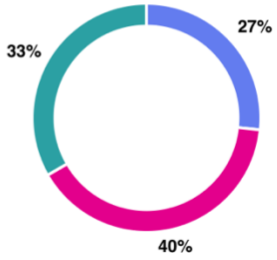


Teen (14-18) and Kids (10-13) responses about pricing (Q=20, n=27)

31. What price would feel acceptable for a child/teen smart ring?

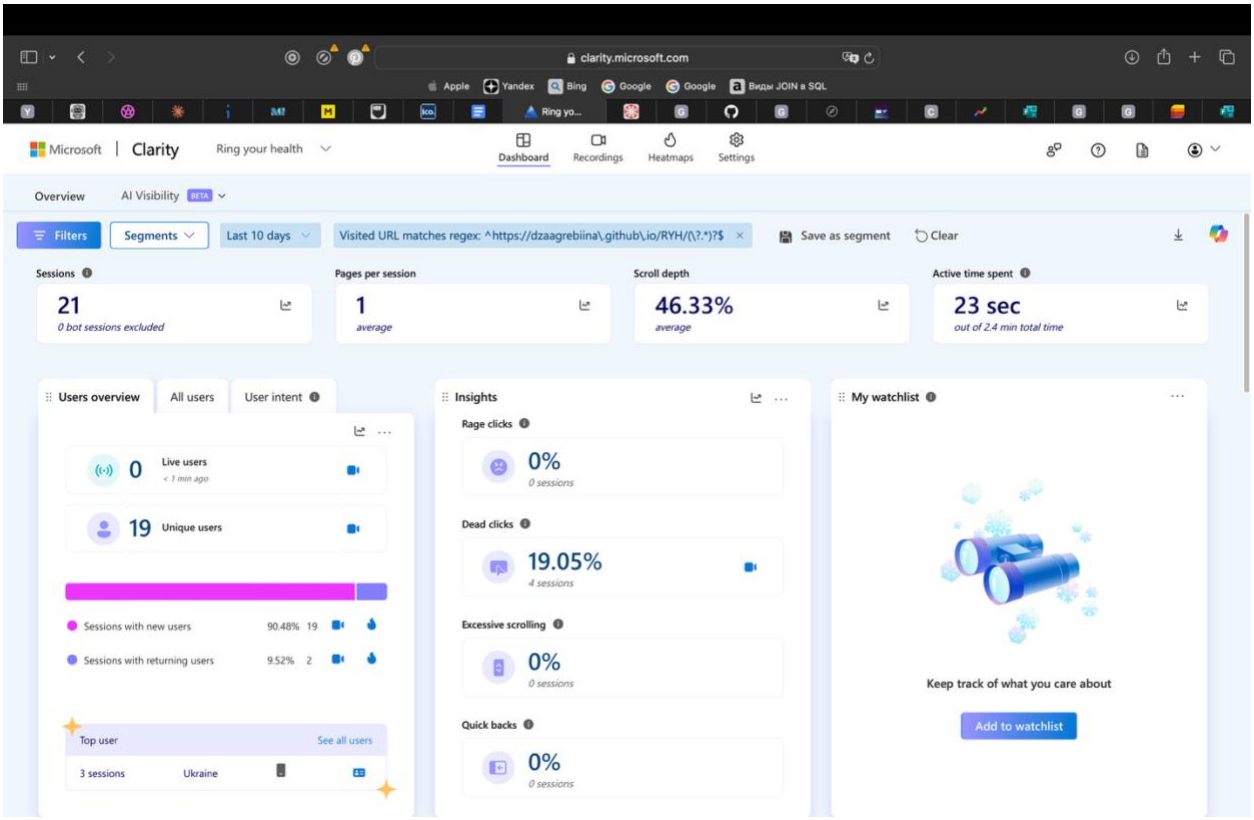
More details

£150–£179	4
£180–£199	6
£200-£220	5
£220+	0

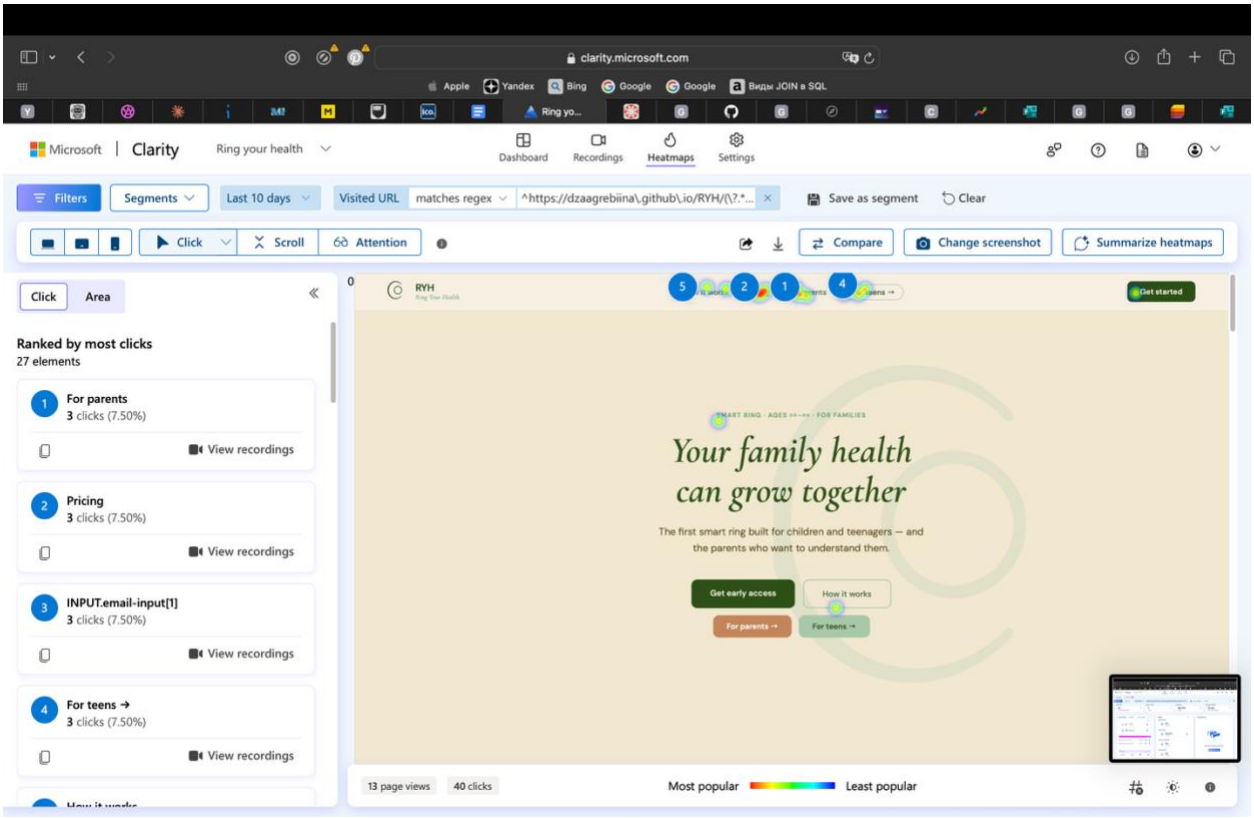


Parents responses on pricing (30-45) Q=31, n =13)

C3 – Microsoft clarity dashboard



C4 – Microsoft Clarity Heatmap

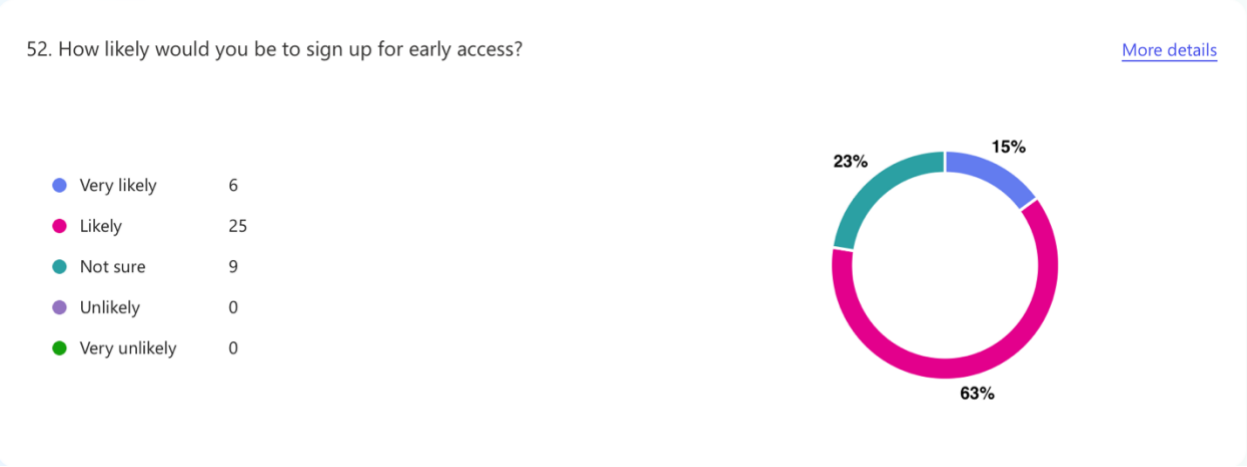


C5 – website navigation



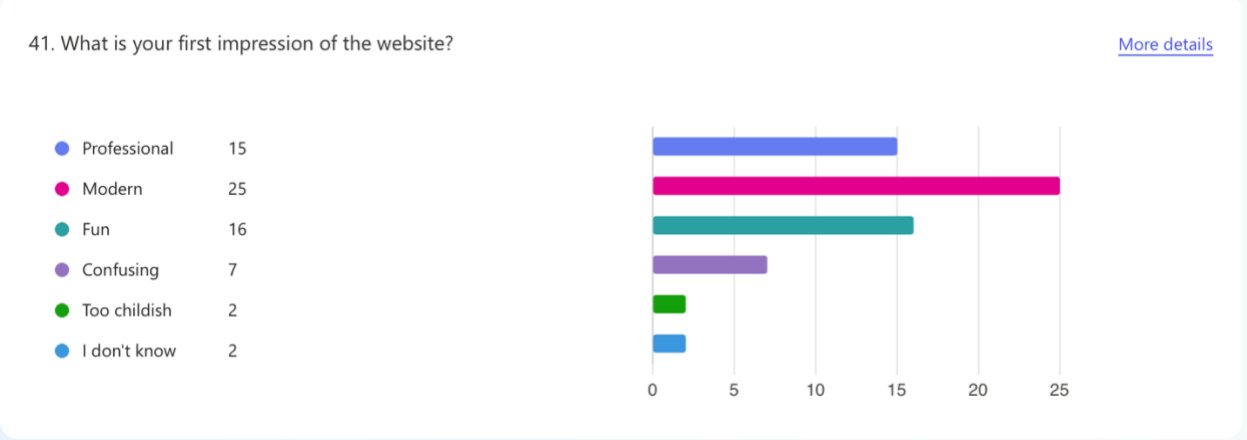
Q=51, n=40

C6 – sign-up for waiting list



Q=52, n=40

C7 website appereance

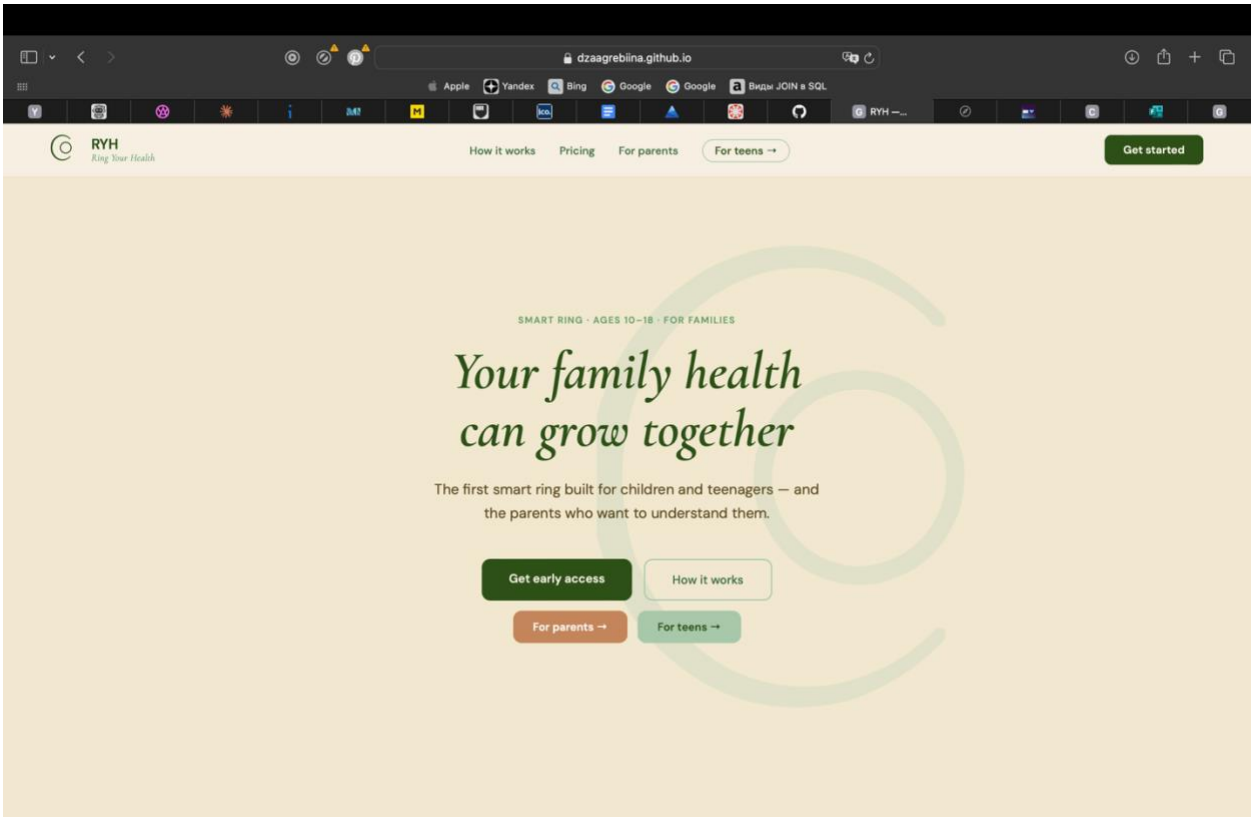


Q=41, n=40

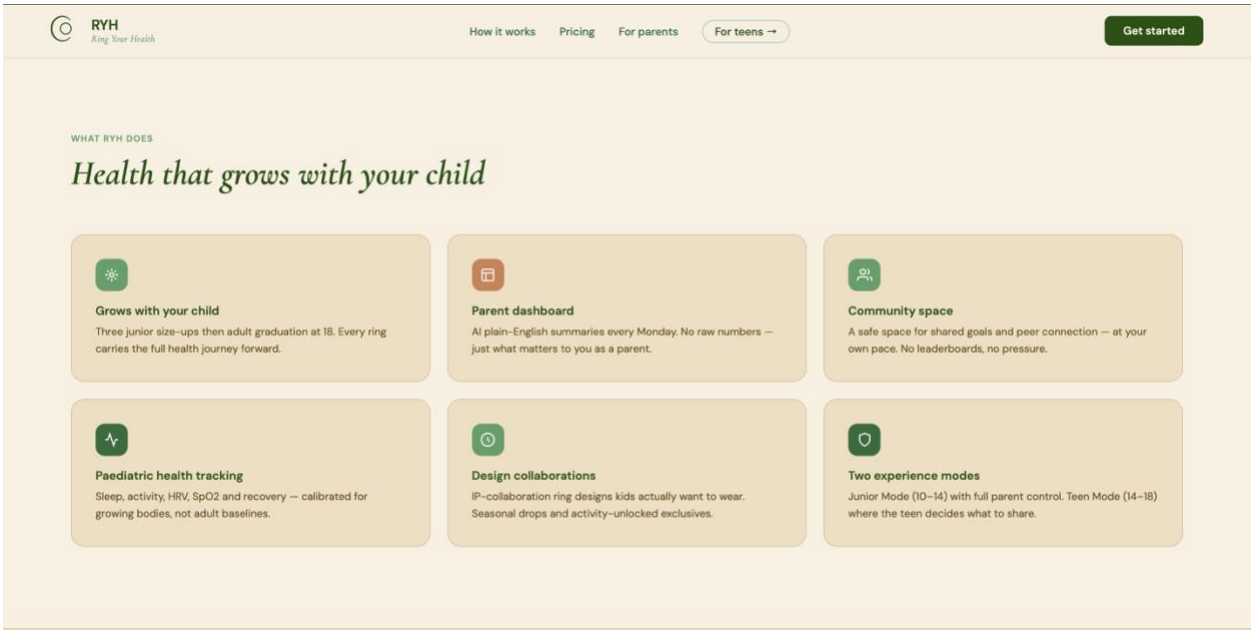
Appendix D - Website

The website link: https://dzaagrebiina.github.io/RYH/#waitlist

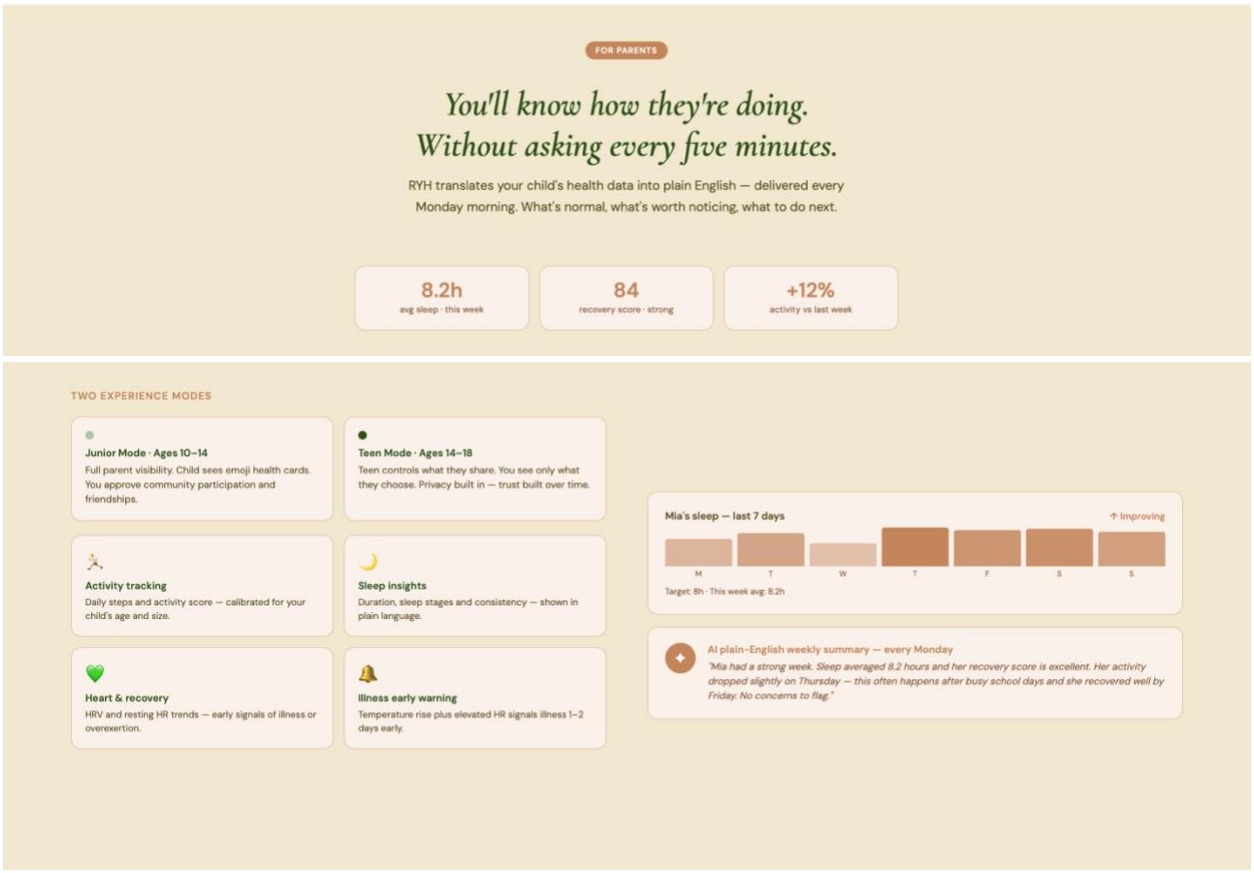
D1 - **RYH Website -Homepage Hero Section**



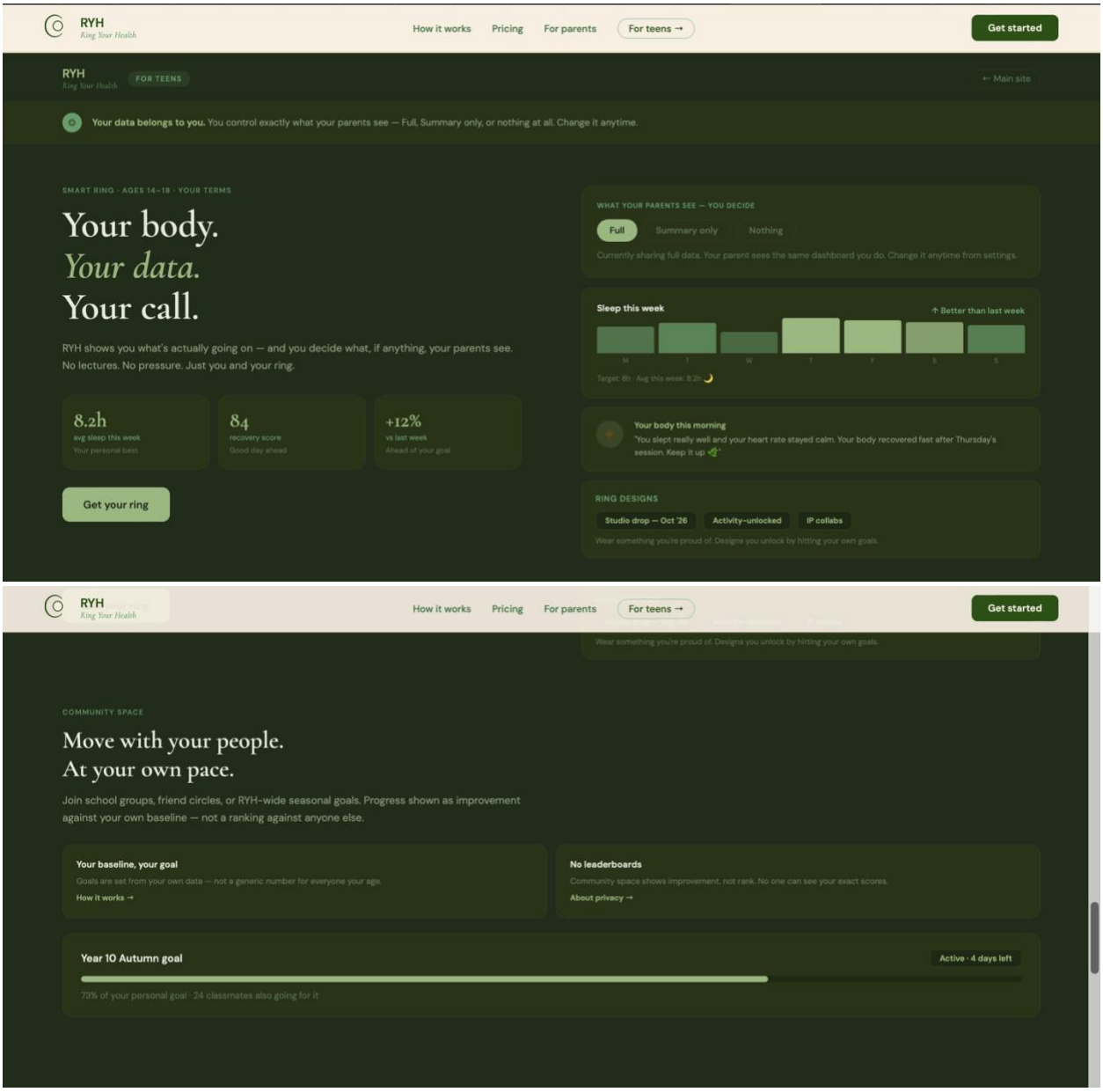
D2 - **RYH Website -Product Features Section**



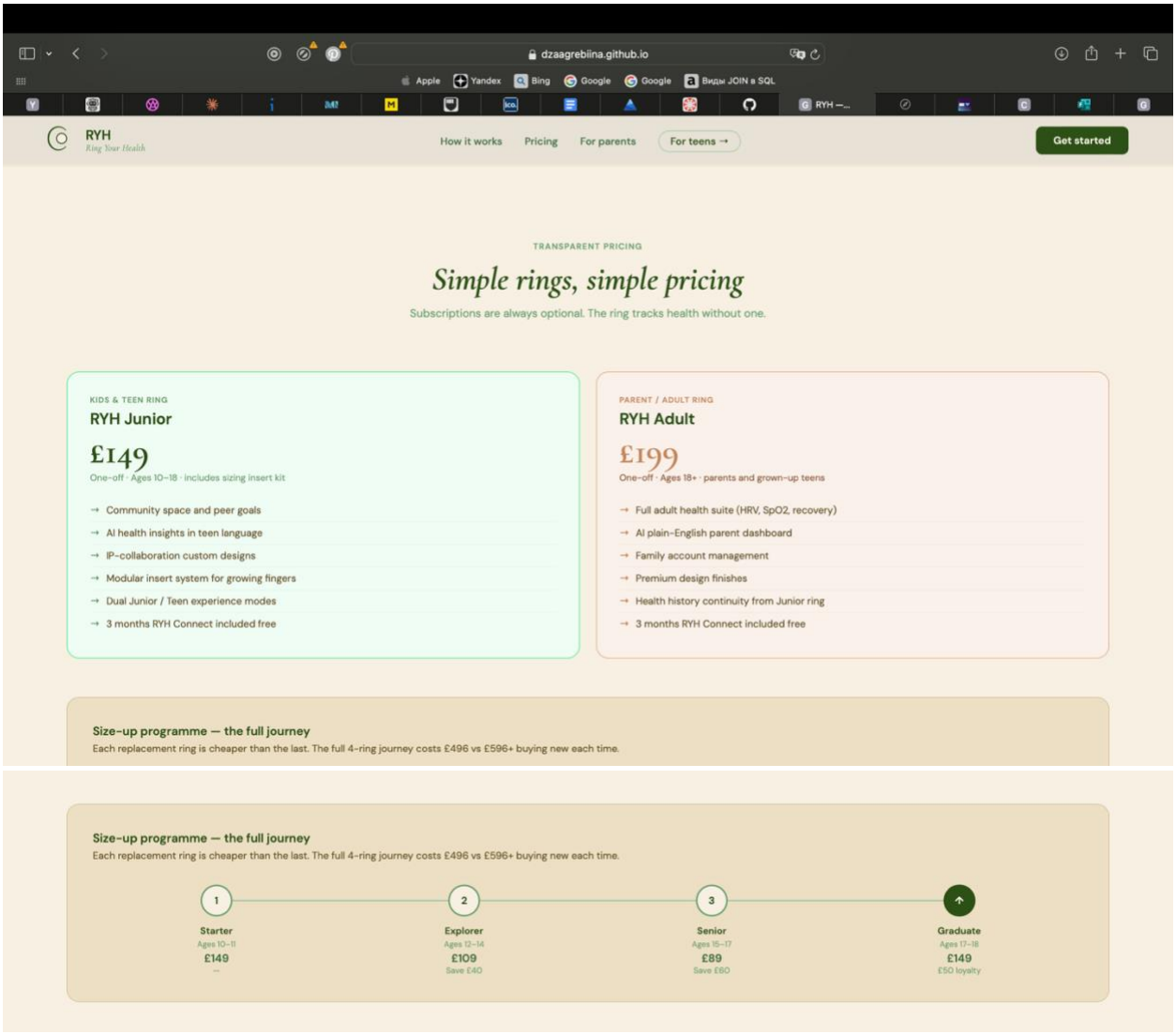
D3 - **RYH Website -Parent Dashboard Section**



D4 - **RYH Website -Teen Section**



D5 - **RYH Website -Pricing Section**



Size-up programme — the full journey

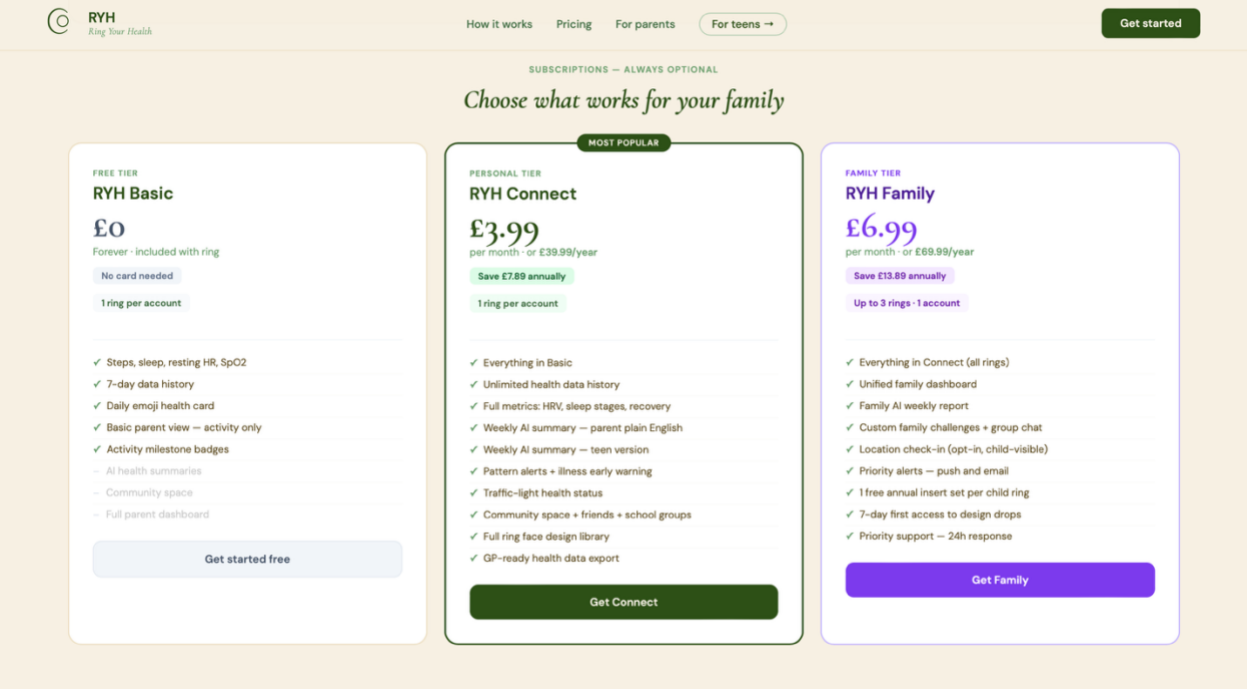
Each replacement ring is cheaper than the last. The full 4-ring journey costs £496 vs £596+ buying new each time.

1 **Starter** Ages 10–11 £149 —

2 **Explorer** Ages 12–14 £109 Save £40

3 **Senior** Ages 15–17 £89 Save £60

↑ **Graduate** Ages 17–18 £149 £50 loyalty



D6 - **RYH Website -Waitlist Section**

